Generative AI at Work

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TASKS VII: The Economic Impacts of AI on Work and Labor Markets

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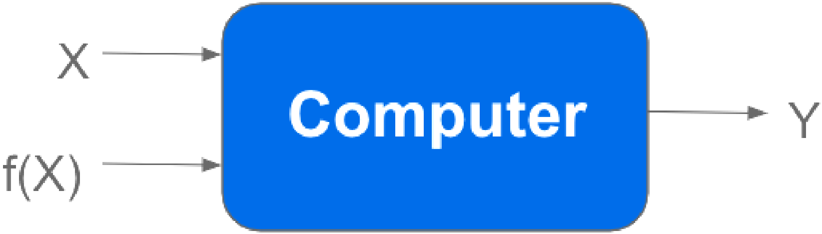
The old: programmable computers

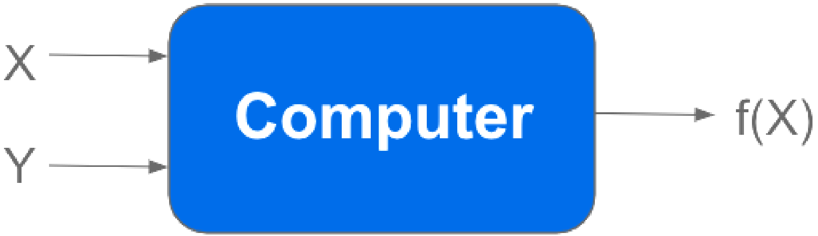
Traditionally, computers require inputs (X) and instructions ($f(X)$) to produce output (Y)



Computers limited to tasks where we know $f(X)$

The new: learn f(X) from data

Machine learning requires inputs (X) and output (Y) to <u>learn</u> instructions ($f(X)$)



- $f(X)$ is an output, not an input
- Polyani's paradox (Autor, 2014)

Generative AI: optimism and apprehension

Generative AI: ML that uses learned patterns to generate new content

- **Optimism**
  - $\rightarrow$ Expands the set of tasks machines can perform

- **Apprehension**
  - $\rightarrow$ Adjustment costs, gulf between lab and real-world scenarios

Generative AI: optimism and apprehension

Generative AI: ML that uses learned patterns to generate new content

- **Optimism**
  - → Expands the set of tasks machines can perform
- **Apprehension**
  - → Adjustment costs, gulf between lab and real-world scenarios
  - → Humans + machines
    - Decision support tools + human seem to do worse than human or algorithm alone (Vaccaro, Almaatouq & Malone, 2024)
  - → Coherent but inaccurate

A paper: Generative AI at Work

Joint work with Erik Brynjolfsson (Stanford) and Danielle Li (MIT)

1. Does access to generative AI suggestions increase productivity?
2. For which workers and why?
3. How does this change the experience of work?

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Setting: Customer support

Technology: AI assistant built on GPT-3

Empirical design: Staggered roll-out at a Fortune 500 software company

A paper: Generative AI at Work

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1. Does access to generative AI suggestions increase productivity?

   15% increase in resolutions per hour

2. For which workers and why?

   Increases are driven by less experienced, less skilled workers

3. How does this change the experience of work? Attrition, customer interactions

Agenda

1. **Setting: customer support**
   - → AI chatbot design and output
2. Data and study design
3. Productivity effects
   - → Aggregate and by skill and experience
   - → Productivity-experience curve
   - → Copying or learning?
   - → Learning what?
4. Experience of work
   - → Customer interactions
   - → Attrition

Customer support is leading AI adoption

Technical feasibility:

- Millions of chats with automatically labeled outcomes

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Business need:

- High turnover
- Large and persistent performance differences

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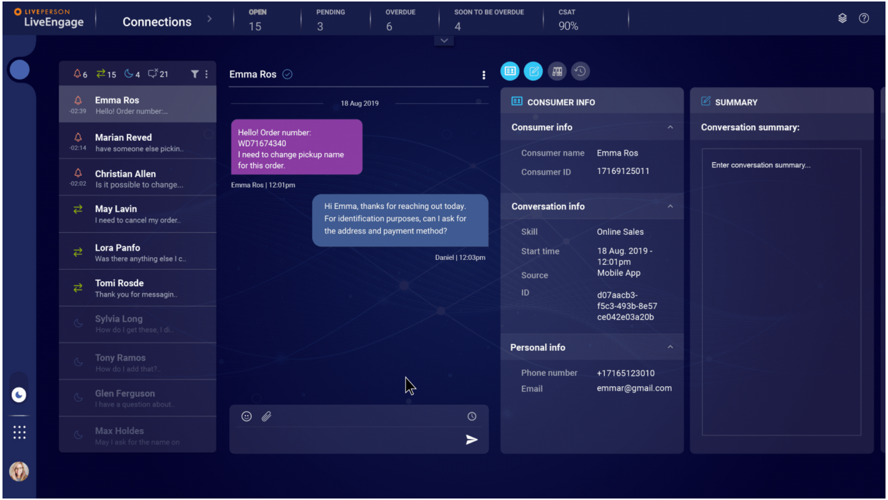
Business need:

- High turnover
- Large and persistent performance differences

Technical and social skills required:

- Computer and accounting knowledge
- Social and emotional intelligence

Customer support



Why is customer support an ML problem?

1. Diagnose the problem
   - → What underlying problems present as “I can’t login"?

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   → What underlying problems present as “I can’t login"?

2. Predict what will resolve it

   → What types of solutions have typically worked for this problem?

3. Recognize what gets people yelled at

   → How to communicate in ways that alleviate customer frustration?

Our AI system

Model design:

1. Start from a large language model from the Generative Pre-Trained Transformer (GPT) family of models (GPT-3)

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   - → Labelled with outcomes: call duration, successfully resolved, customer satisfaction, “high performer” or not

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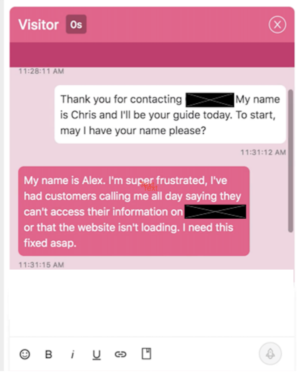
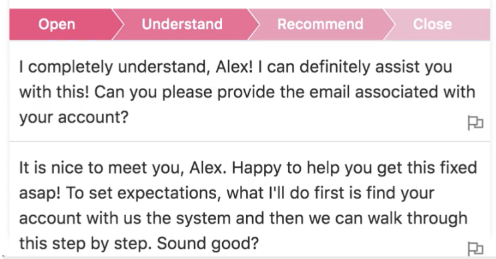
System function:

- Real-time text suggestions and links to technical material
- Worker retains full discretion

AI tool provides suggestions to the agent





- Recommendations based on responses that are most correlated with successful outcomes
- In this case: establishing a friendly, reassuring rapport.

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Our data

Sample

- 3 million customer service chats between 2020-2021
- 5,000 agents; 1,500 treated
- 5 firms (data firm and 4 outsourcing firms)

Chat level information

- Customer and agent text
- Adherence to AI suggestions
- Chat duration

Agent-month level information

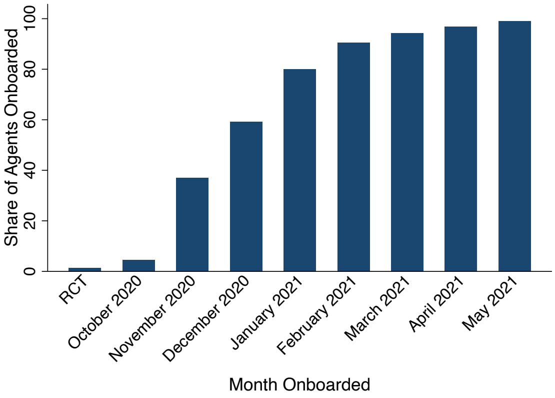
- Average call resolution rate
- Average surveyed customer satisfaction

Summary stats Balance table

Empirical strategy: staggered rollout of AI

Rollout shaped by two constraints:

- Scheduling
  - $\rightarrow$ Limited training personnel $\Longrightarrow$ long rollout
  - $\rightarrow$ Scheduling constraints $\Longrightarrow$ agent-level rollout
- Budget
  - $\rightarrow$ Small RCT before deployment
  - $\rightarrow$ Fixed dollar budget



Empirical strategy: staggered rollout of AI

Study Design

$$\underbrace{y_{it}}_{\text{Resolutions per hour}} = \alpha_i + \delta_t + \sum_{j \neq -1}^{J} \beta_j (AI_{it} \times \mathbb{1}[t = j]) + \gamma X_{it} + \epsilon_{it}$$

- AI rolled out over six months at the agent level
- AI_{it} indicates agent i has access to AI assistance at time t
- X_{it} months of agent tenure
- Use estimators robust to differential timing
- Data available at the agent-month level

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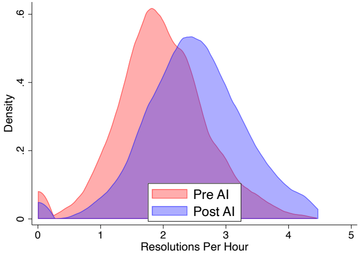
Gains are evident in the raw data



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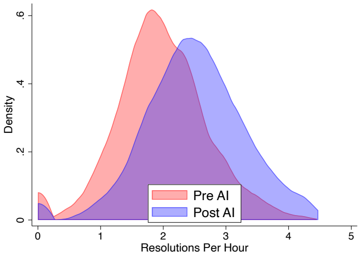

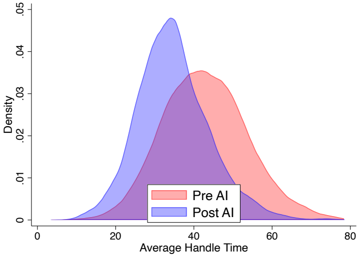

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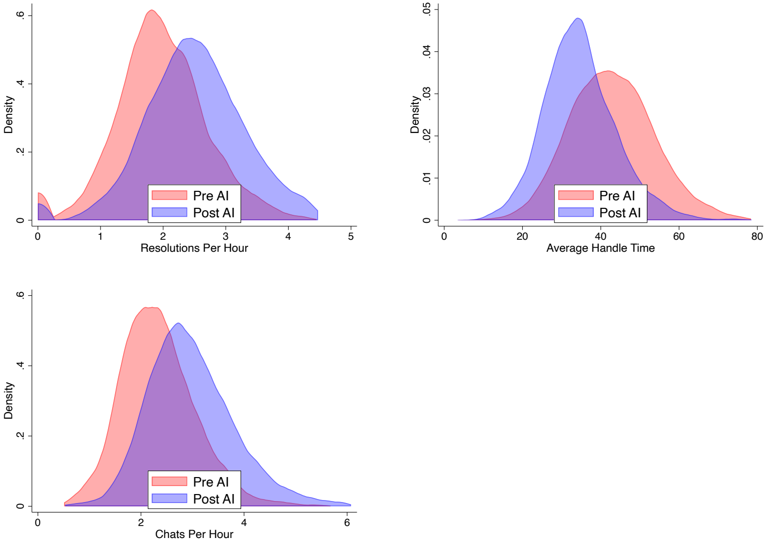

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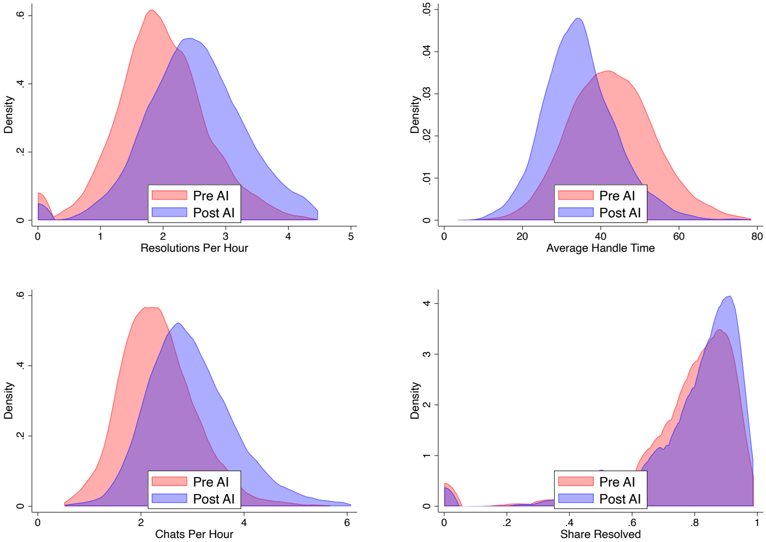
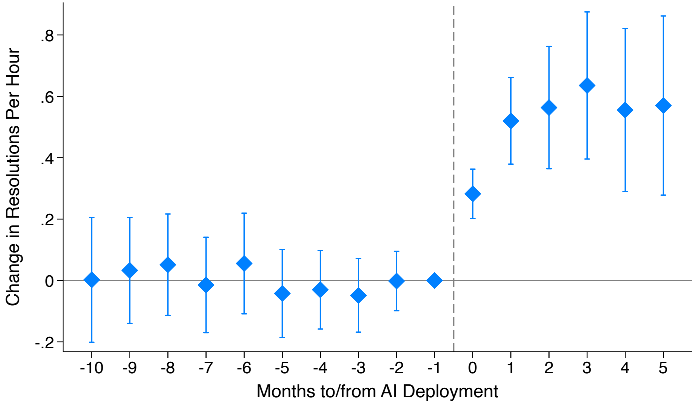
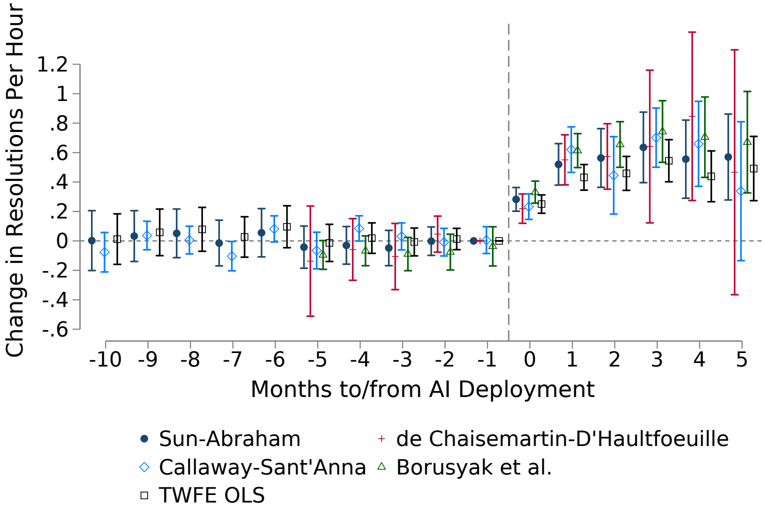


On average, a 15 percent productivity improvement

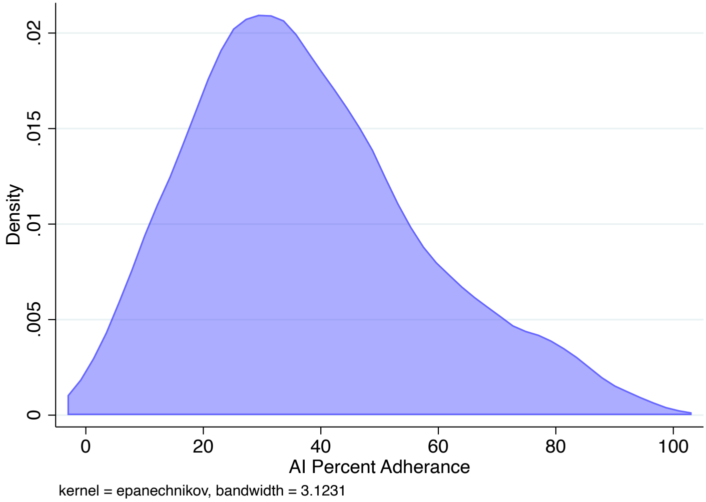


Difference-in-differences table | AI adherence

Similar results across estimators

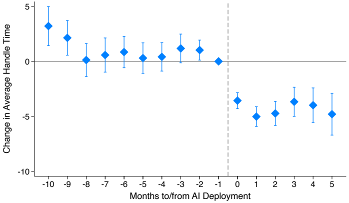


35% of AI suggestions followed



Event studies, additional outcomes

A. Average Handle Time



B. Chats Per Hour

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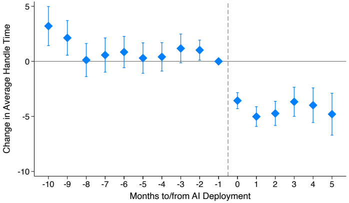

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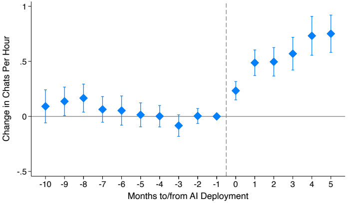

C. Resolution Rate

D. Customer Satisfaction (NPS)

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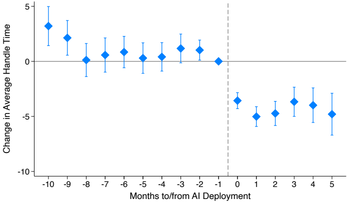

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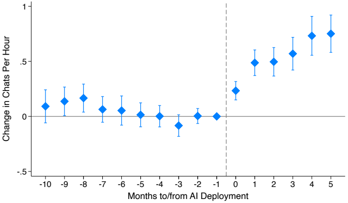

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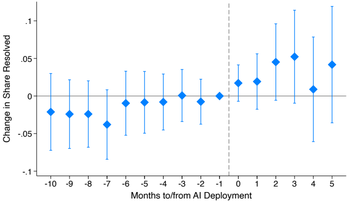

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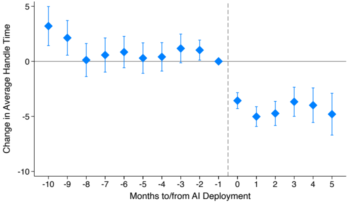

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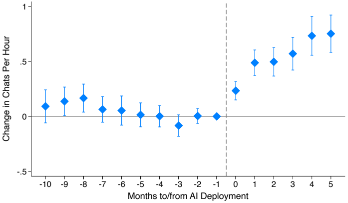

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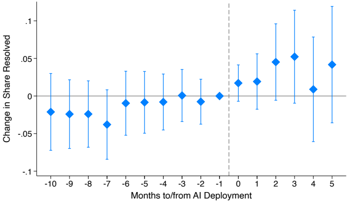

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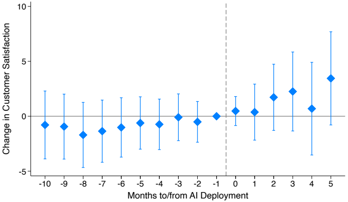

How does the value of AI assistance vary across workers?

Higher ability workers use AI more effectively

- Prior evidence of skill bias in IT (Bartel et al., 2007)
- AI requires discretion

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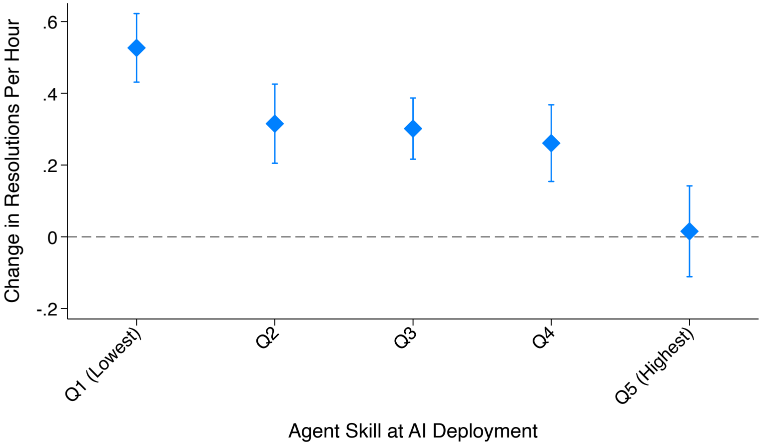
We measure worker skill and experience

- Skill: pre-AI index of all outcomes
- Experience: pre-AI tenure with the firm

Highest returns for the lowest skill agents



▸ Tenure

Additional outcomes by worker skill

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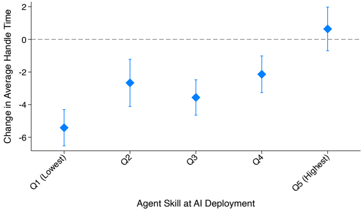
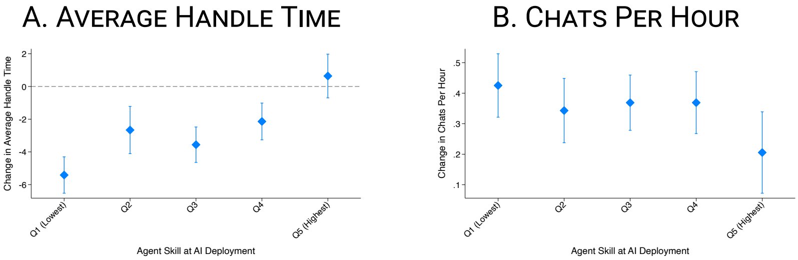


B. CHATS PER HOUR

Additional outcomes by worker skill



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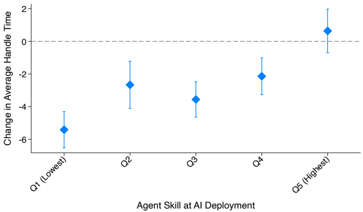

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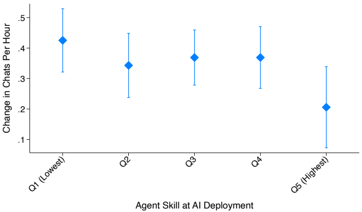

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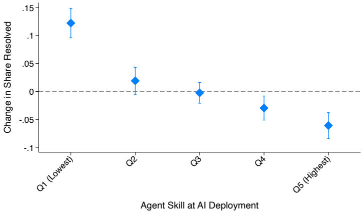

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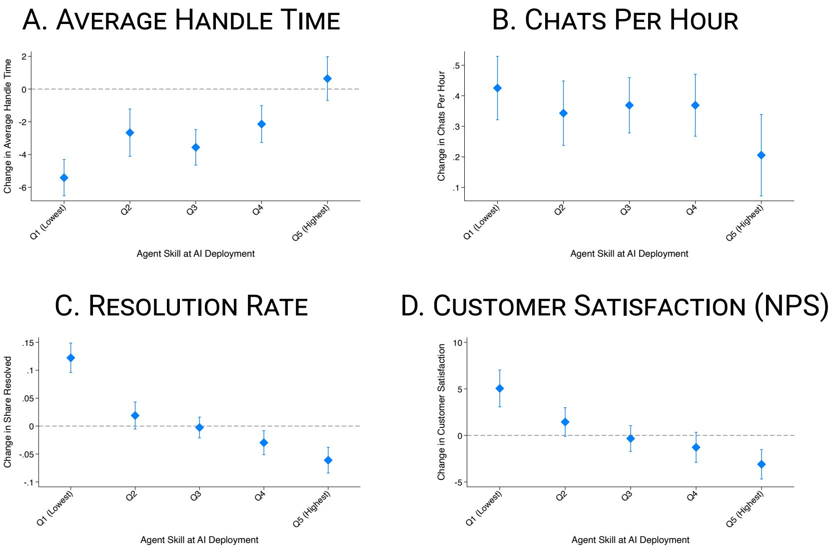

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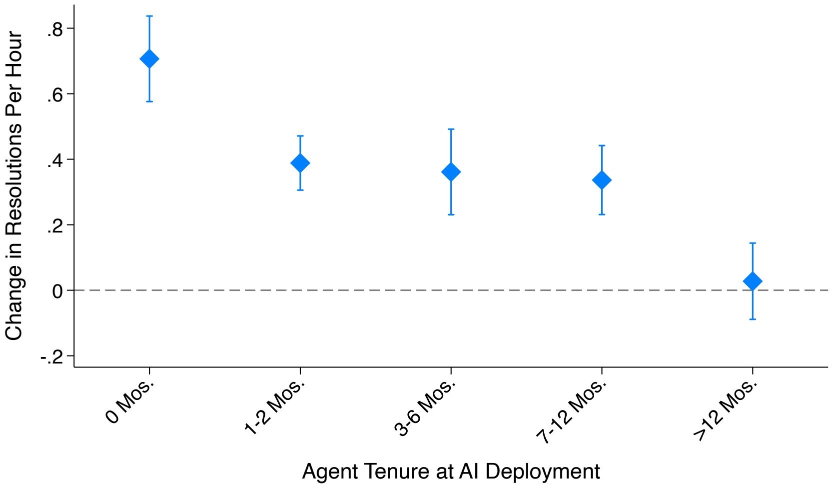

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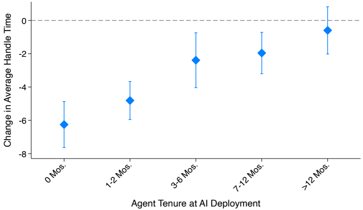

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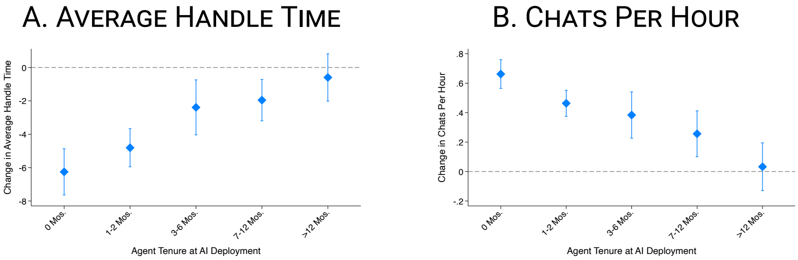

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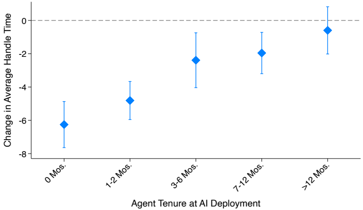

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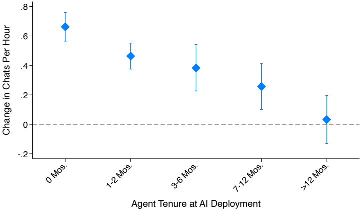

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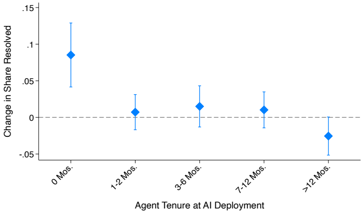

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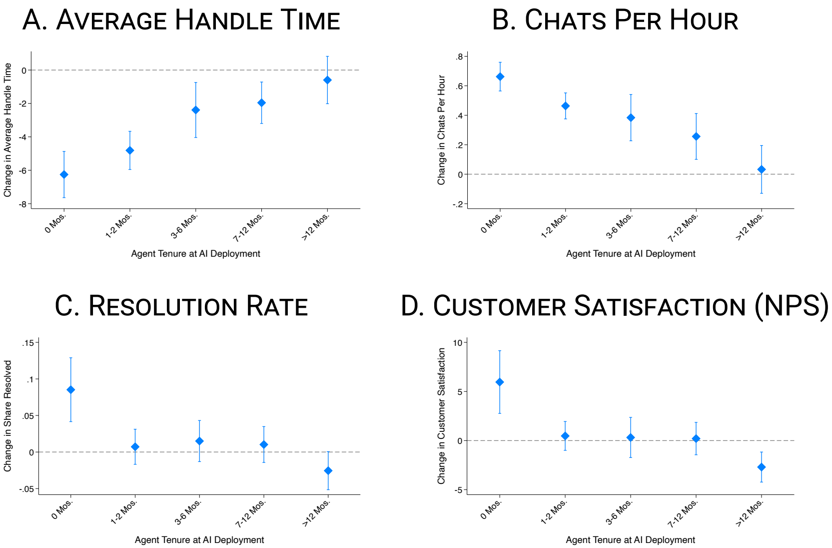
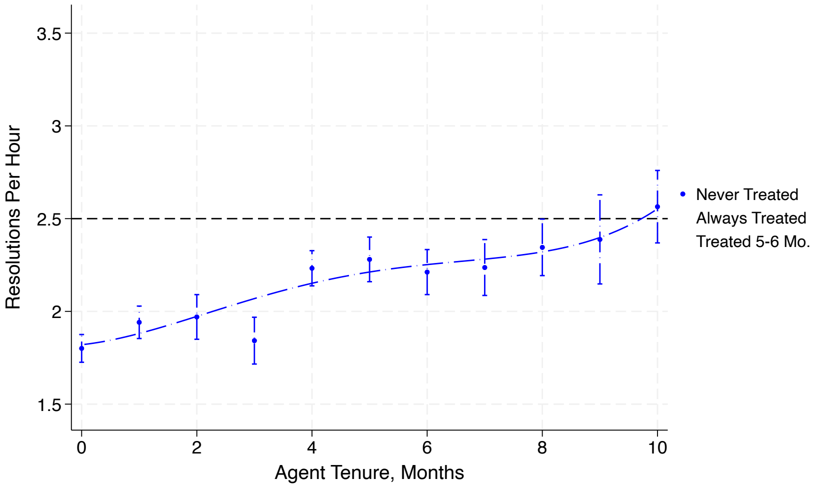


AI assistance helps newer agents “catch up”



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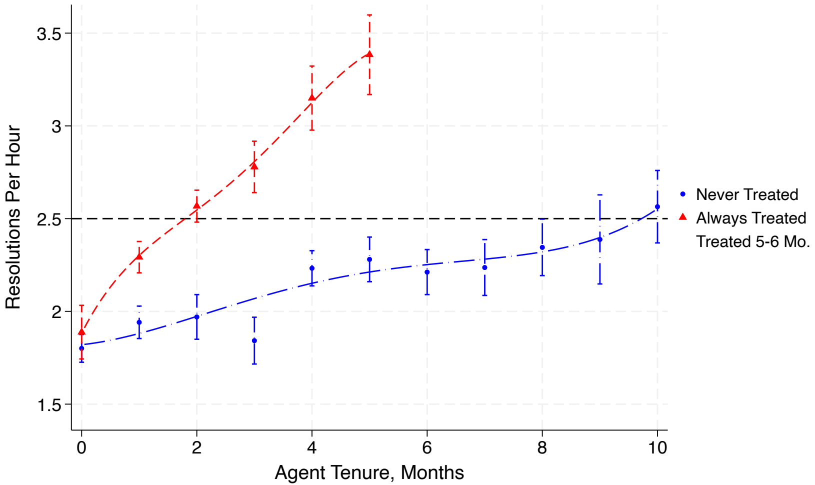

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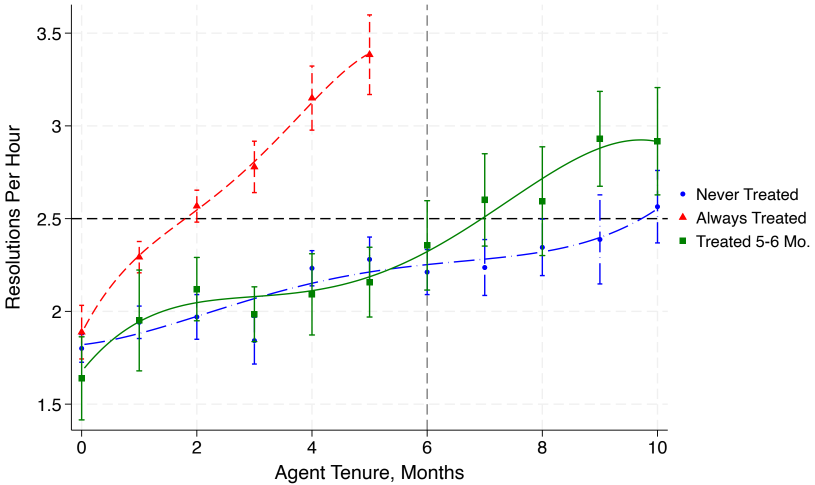

Robustness checks

- Instrument individual adoption with earliest date of team adoption ▸ IV estimates
- Alternative diff-in-diff specifications ▸ Alternative specifications
- Analyze randomized control trial ▸ Randomized control trial
- Vary the level of clustering ▸ Clustering
- Weight by the number of chats ▸ Weighting
- Mean reversion ▸ mean reversion?

Testing for worker learning

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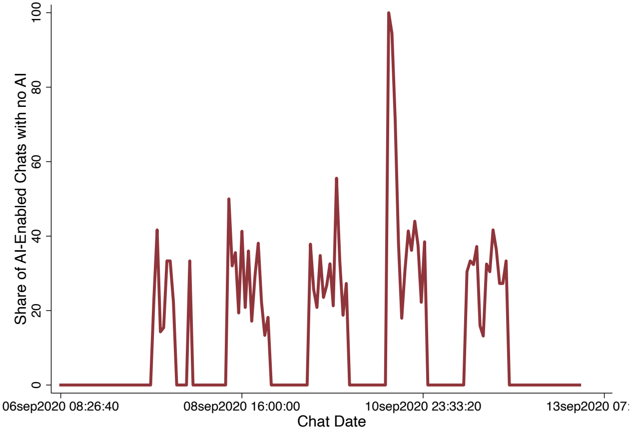
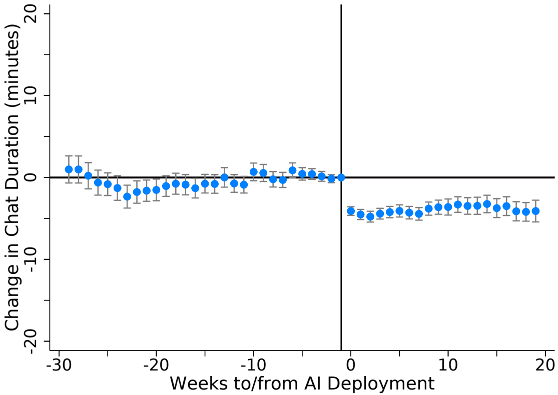
Occasional periods of AI outages



AI-exposed workers faster during AI outages after exposure

a. AI Working



b. AI Outage

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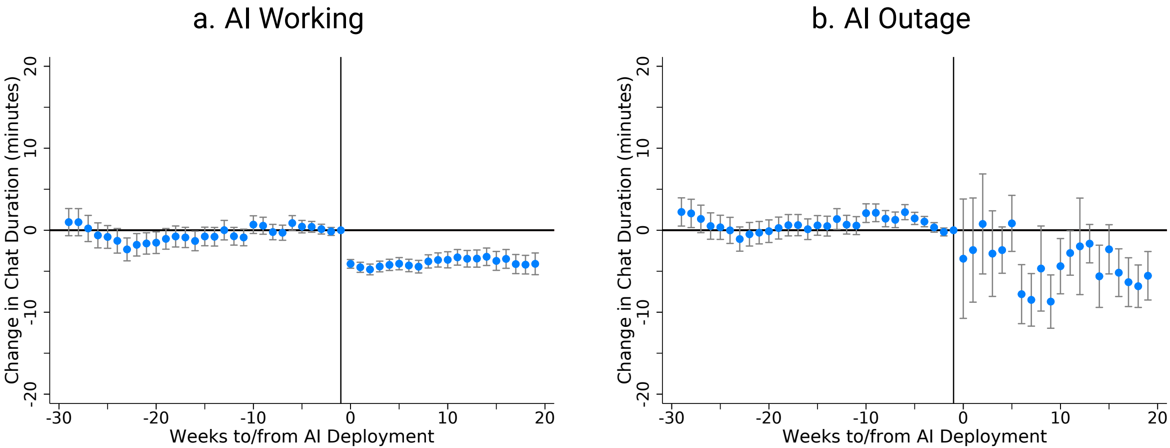
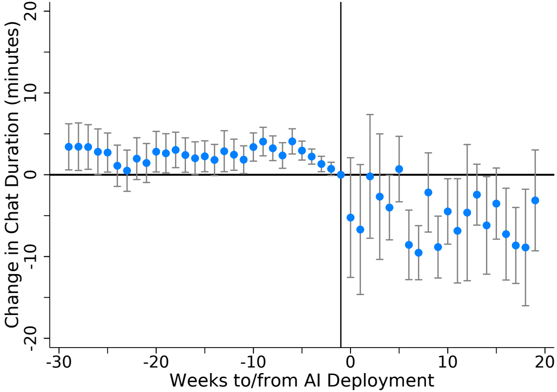


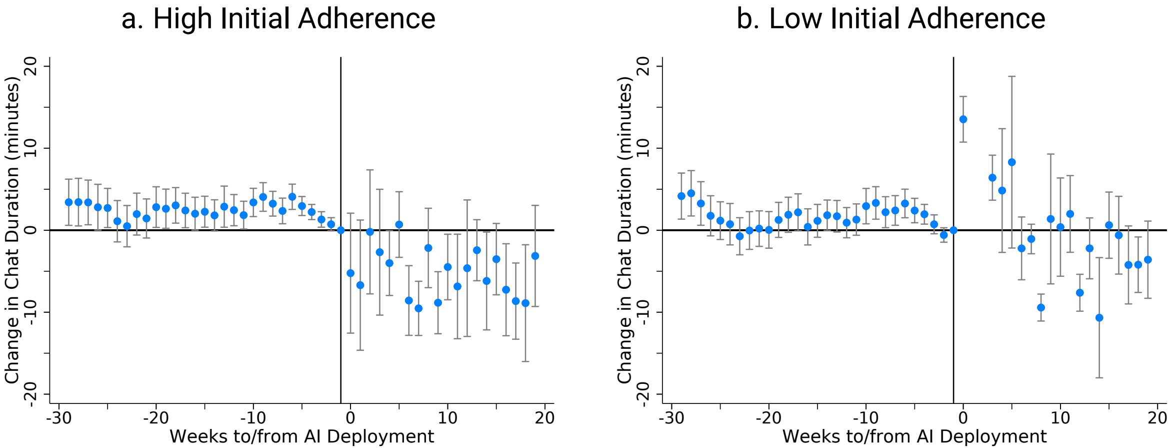
Intensive AI users are faster during outages

a. High Initial Adherence



b. Low Initial Adherence

Intensive AI users are faster during outages



Assessing mechanism

1. AI learns patterns associated with success

   Challenge: patterns are not human-interpretable

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   $\Longrightarrow$ Assess chat text

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Evidence on mechanism:

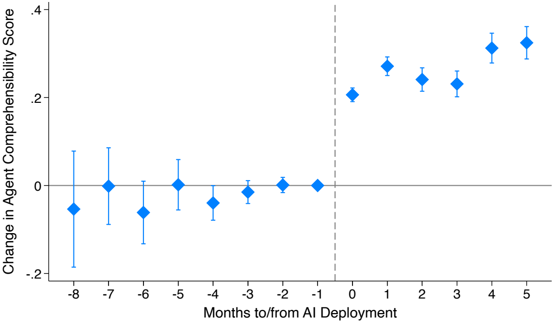
- Analyze impacts on English proficiency
- Compare high and low skill workers' language convergence
- Impacts by problem topic

English language fluency

- Good call center agents communicate with comprehensible and native-sounding English
- We use the chat transcripts to measure;
  1. **Comprehensibility** - a score ranging from 1 (“very difficult to comprehend”) to 5 (“very fluent and easily understandable, with no significant errors”)
  2. **Native fluency** - Interagency Language Roundtable “functionally native” proficiency standard score from 1 to 5
- Transcripts scored by Gemini, a LLM, and LLM scores are human-validated
- Does access to AI impact agent comprehensibility or native fluency?

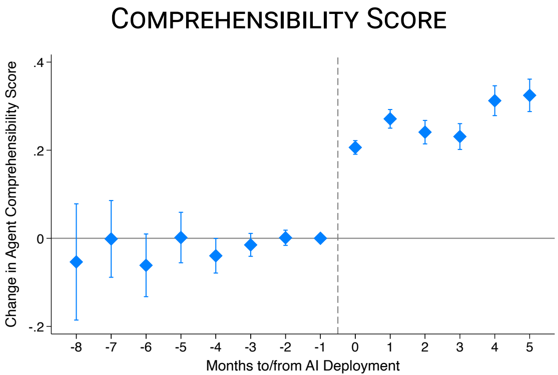
AI access increases English fluency

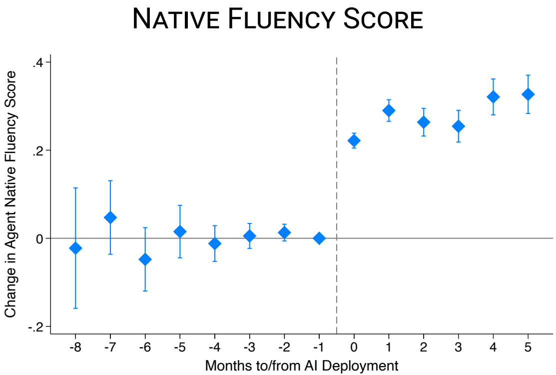
Comprehensibility Score



Native Fluency Score

AI access increases English fluency

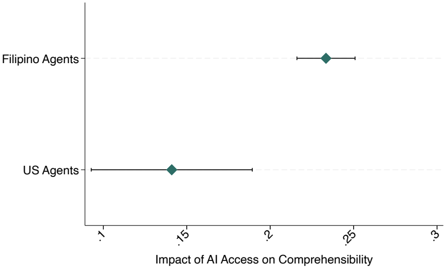




- 5% increase in comprehensibility score
- 6% increase in native fluency score

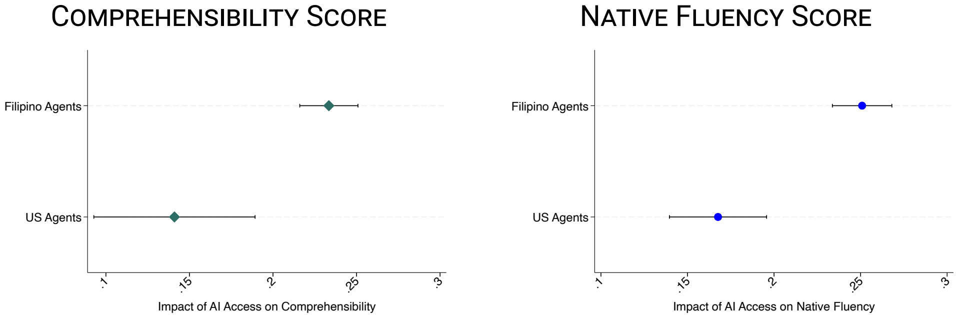
Filipino agents linguistic improvements double the U.S workers

Comprehensibility Score



Native Fluency Score

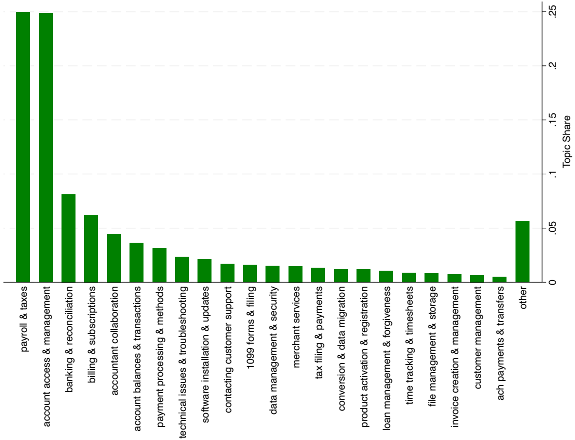
Filipino agents linguistic improvements double the U.S workers



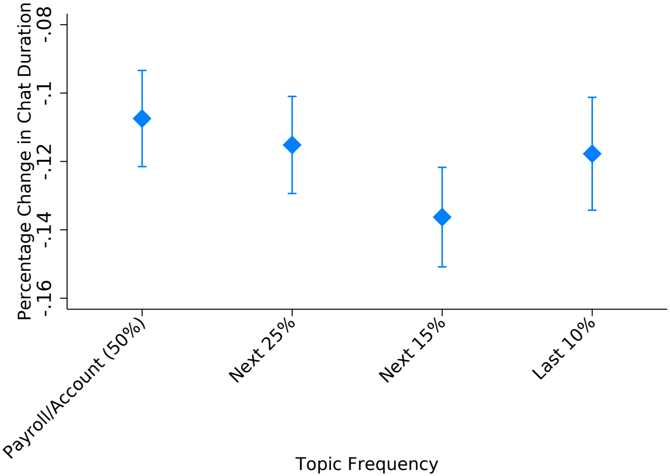
- **Comprehensibility:** US +3.5%; Philippines +5.7%
- **Native Fluency:** US +2.9%; Philippines +5.3%
- Customer engagement occurs across a wide range of jobs

Identify the chat topic

- Randomly sample 5,000 chats
- Ask for a short description of chat topic
- Cluster descriptions into 50 topics
- Classify all conversations into topic
- Human validation



Impacts of AI along topic distribution



Conclusion

Evidence from contact centers

- Clear, quantifiable outcomes
- Mix of repetitive tasks and less routine tasks

Raises further questions:

- **Employment:** Corresponds to 12% fewer worker-hours
  - $\rightarrow$ Where will those jobs be located?
- **Wages:** Nature of the existing contract structure
  - $\rightarrow$ Workers: performance-based pay
  - $\rightarrow$ Outsourcing firms: contracts in hours of work

Questions: lraymond@mit.edu

Appendix I

Sample Summary Statistics

Variable	All	Never Treated	Treated, Pre	Treated, Post
Chats	3,007,501	945,954	882,105	1,180,446
Agents	5,179	3,523	1,341	1,636
Number of Teams	133	111	80	81
Share US Agents	.11	.15	.081	.072
Distinct Locations	25	25	18	17
Average Chats per Month	127	83	147	188
Average Handle Time (Min)	41	43	43	35
St. Average Handle Time (Min)	23	24	24	22
Resolution Rate	.82	.78	.82	.84
Resolutions Per Hour	2.1	1.7	2	2.5
Customer Satisfaction (NPS)	79	78	80	80

Table: Sample Summary Statistics

On average, a 15 percent productivity improvement

VARIABLES	(1) Resolutions/Hr	(2) Resolutions/Hr	(3) Resolutions/Hr
Post AI X Ever Treated	0.469***	0.371***	0.301***
	(0.0325)	(0.0318)	(0.0329)
Ever Treated	0.110**		
	(0.0440)		
Observations	13,192	12,295	12,295
R-squared	0.249	0.562	0.575
Year Month FE	Yes	Yes	Yes
Location FE	Yes	Yes	Yes
Agent FE	-	Yes	Yes
Agent Tenure FE	-	-	Yes
DV Mean	2.123	2.176	2.176

Robust standard errors in parentheses

*** $p<0.01$, ** $p<0.05$, * $p<0.10$

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Team Company IV

VARIABLES	(1) Individual Treatment	(2) Res./Hr	(3) AHT	(4) Chats/Hr	(5) Res. Rate	(6) NPS
Earliest Team Treatment	0.311***					
	(0.0163)					
Post AI X Individual Treatment		0.550***	-3.622**	0.412***	0.0740***	-0.518
		(0.0943)	(1.444)	(0.0791)	(0.0184)	(0.946)
Observations	21,839	12,295	21,839	21,839	12,295	12,541
R-squared	0.813	0.177	0.102	0.115	0.007	0.007
Year Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Agent FE	Yes	Yes	Yes	Yes	Yes	Yes
Agent Tenure FE	Yes	Yes	Yes	Yes	Yes	Yes
F-statistic	365.7					
Number of agent_id		2,163	3,633	3,633	2,163	2,214

Robust standard errors in parentheses

*** $p<0.01$, ** $p<0.05$, * $p<0.10$

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Alternative Specifications

VARIABLES	(1) Resolutions/Hr	(2) Resolutions/Hr	(3) Resolutions/Hr	(4) Resolutions/Hr
Post AI X Ever Treated	0.469***	0.371***	0.301***	0.241***
	(0.0325)	(0.0318)	(0.0329)	(0.0402)
Ever Treated	0.110**			
	(0.0440)			
Observations	13,192	12,295	12,295	11,855
R-squared	0.249	0.562	0.575	0.644
Year Month FE	Yes	Yes	Yes	Yes
Location FE	Yes	Yes	Yes	Yes
Agent FE	-	Yes	Yes	Yes
Agent Tenure FE	-	-	Yes	Yes
Team X Month FE	-	-	-	Yes
DV Mean	2.123	2.176	2.176	2.181

Robust standard errors in parentheses

*** $p<0.01$, ** $p<0.05$, * $p<0.10$

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Randomized control trial

	(1)	(2)	(3)	(4)	(5)
VARIABLES	Res./Hr	AHT	Chats/Hr	Res. Rate	NPS
Post AI X Ever Treated	0.202**	-3.713***	0.105	0.0169	-1.393
	(0.0850)	(1.045)	(0.0718)	(0.0246)	(1.529)
Observations	6,998	15,100	15,100	6,998	7,176
R-squared	0.568	0.574	0.566	0.383	0.517
Year Month FE	Yes	Yes	Yes	Yes	Yes
Agent FE	Yes	Yes	Yes	Yes	Yes
Agent Tenure FE	Yes	Yes	Yes	Yes	Yes
DV Mean	1.956	43	2.378	0.808	79.68

Robust standard errors in parentheses

*** $p<0.01$, ** $p<0.05$, * $p<0.10$

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Robustness to alternative clustering

	(1)	(2)	(3)
VARIABLES	Res./Hr	Res./Hr	Res./Hr
Post AI X Ever Treated	0.301***	0.301***	0.301***
	(0.0329)	(0.0455)	(0.0498)
Observations	12,295	12,295	12,295
R-squared	0.575	0.575	0.575
Year Month FE	Yes	Yes	Yes
Agent FE	Yes	Yes	Yes
Agent Tenure FE	Yes	Yes	Yes
DV Mean	2.176	2.176	2.176

Robust standard errors in parentheses

*** $p<0.01$, ** $p<0.05$, * $p<0.10$

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Weighting by number of chats

VARIABLES	(1) Res./Hour	(2) AHT	(3) Chats/Hr	(4) Res. Rate	(5) NPS
Post AI X Ever Treated	0.252***	-3.100***	0.295***	0.00325	-0.119
	(0.0318)	(0.315)	(0.0263)	(0.00725)	(0.524)
Observations	12,295	21,839	21,839	12,295	12,541
R-squared	0.650	0.756	0.721	0.465	0.526
Year Month FE	Yes	Yes	Yes	Yes	Yes
Agent FE	Yes	Yes	Yes	Yes	Yes
Agent Tenure FE	Yes	Yes	Yes	Yes	Yes
DV Mean	2.457	38.66	2.886	0.839	79.59

Robust standard errors in parentheses

*** $p<0.01$, ** $p<0.05$, * $p<0.10$

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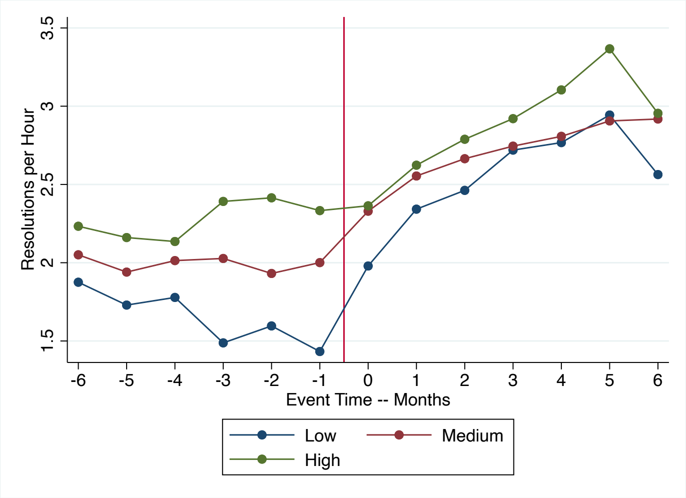
Evidence of mean reversion?



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